Web Basics

HTML — Structure / layout

CSS (Cascading style sheets) — Style

JavaScript — Logic

It provide functionality

HTML (Hyper text Markup language)

HTML is the code that is used to structure a web page and its content.

The component used to design the structure of webfiles are called HTML tags.

HTML Tag

A container for some content or other HTML tags

<P> This is a paragraph </P>

content

Element

Basic HTML Page

```
<!DOCTYPE html>          tells browser you are using HTML5

<html>                   root of an html document
  <head>                 container for metadata
    <title> My first Page </title>    page title
  </head>
  <body>                 Contains all data rendered by the
    <p> hello world </p>   paragraph tag

  </body>

</html>
```

not display on page

Quick Pt

- Html tag is parent of head & body tag
- Most html elements have opening & closing tags with content in b/w

  `<br>` — there is no opening or closing

- Just like `<br>` tag there are tags with no opening or closing
- We can use inspect element / view page source to edit html

Comments in HTML

This is a part of code that should not be parsed.

```
<!-- This is an HTML Comment -->
```

HTML is NOT case Sensitive

<html> = <HTML>
 = <P>
<head> = <HEAD>
<body> = <BODY>

Basic HTML Tags

① HTML Attributes – Attributes are used to add more information to the tag

<html lang = "en">
↳ Attributes

② Heading Tag – used to display headings in HTML

h1 – most important
h2
h3
h4
h5
h6 – least important

font size decreases.

* never use heading-Tag as a big heading or small heading. It is only used for showing importance not for font size.

③ Paragraph Tag – used to add paragraphs in HTML

 This is a sample paragraph

④ Anchor Tag – used to add links to your page

 Google

⑤ Image Tag – used to add images to your page

```
<img src="/image.png" alt="Random Image"/>
```

↓

this works only when the image is unable to load from the "img src" link.

– Height of an image is set by

```
<img src="link" alt="notebook" height="100"/>
```

* It is recommended to use CSS for setting height and width.

⑥ Br Tag – used to add next line to your page

```
<br>
```

⑦ Bold, Italic & Underline Tags

└ used to highlight text in your page

```
<b> Bold </b>
<i> Italic </i>
<u> underline </u>
```

⑧ Big & Small Tags

└ used to display big & small text on your page

```
<big> Big </big>

<small> Small </small>
```

(9) Hr Tag — used to display a horizontal ruler, used to separate content

<hr>

(10) Subscript & Superscript Tag

_{Subscript}

H_2O

^{Superscript}

$A^n + B$

(11) Pre Tag — Used to display text as it is (without ignoring spaces & next line)

<Pre> This
is a sample
text.
</Pre>

Page layout Techniques

using Semantic tags for layout

<header>
<main>
<footer>

```
<body>
    <header>
        <h1> experience </h1>
    </header>
</body>
</html>
```

```
<body>
    <header>

    </header>

    <main>

    </main>

    <footer>

    </footer>
</body>
```

Inside main Tag

Section Tag — for Section on your page
<Section>

Article Tag — for an article on your page
<article>

Aside Tag — for content aside main content eg - ads
<aside>

```
<main>
    <section> education : NSIT (COE) | School (JMV) </section>
    <section> experiences : Microsoft | DRDO Intern | MS Intern </section>
    <article> my Story : </article>
    <aside> this is an ad </aside>
```

Rewriting Anchor Tag

```
<a href = "https://google.com" target = "_main" > Google </a>
```

this will open a new tab

```
<a href = "https://google.com" ><img src = "link" > </a>
```

this will make picture clickable

Div Tag – is a container used for other HTML elements

- Non-Semantic Tag
- Div is on Block Elements

```
<div>
    <p> hello world </p>
    <p> namaste </p>
    <p> hi </p>
</div>

<div>
    <p> bye </p>
    <p> goodbye </p>
    <p> ciao </p>
</div>
```

List of Block element

<address>
<article>
<aside>
<blockquote>
<canvas>
<dd>

<dt>

<fieldset>
<figcaption>
<figure>
<footer>
<form>
<h1> - <h6>
<header>
<hr>

<main>

<nav>
<noscript>

<p>
<pre>
<section>
<table>
<tfoot>

<video>

Span Tag - Span is also a container used for other HTML elements

* Inline Element (taken width as per size)

 bye

List of In-line Tags

<a>
<abbr>
<acronym>
<b>
<bdo>
<big>

<button>
<cite>

<code>
<dfn>

<i>
<img>
<input>
<kbd>
<label>
<map>
<object>
<tt>
<var>

<output>
<q>
<samp>
<script>
<select>
<small>

<sub>
<sup>
<textarea>
<time>

Lists in HTML

- Lists are used to represent real life list data.

unordered

```
<ul>
  <li> Apple </li>
  <li> Mango </li>
</ul>
```

ordered list

```
<ol>
  <li> Apple </li>
  <li> Mango </li>
</ol>
```

⇒ This will give n.o's in list

1. Apple
2. Mango

```
<main>
  <ul>
    <li> Apple </li>
    <li> Mango </li>
    <li> Litchi </li>
  </ul>
</main>
```

Tables in HTML

- Tables are used to represent real life table data.

`<tr>` – used to display table row
`<td>` – used to display table data
`<th>` – used to display table header

```
<table> ——→ <caption> Student Data </caption>
  <tr>
    <th> Name </td>
    <th> Roll No </th>
  </tr>
  <tr>
    <td> Ankit </td>
    <td> 12345 </td>
  </tr>
```

</table>

thead & tbody in Tables –

<thead> – to wrap table head
<tbody> – to wrap table body

<th colspan="2"> – this will take the space of two columns of a header

Form in HTML – forms are used to collect data from the user

<form>
form content
</form>

Action in Form – Action attribute is used to define what action needs to be performed when a form is submitted

<form action="/action.php">

</form>

Input
<input type="text" placeholder="Enter Name">

Label
<label for="id1">
<input type="radio" value="does X" name="does" id="id1">
</label>

```
<label for="id2">
    <input type="radio" value="class x" name="class
```

class & Id

```
<div id="id1" class="group1">
</div>

<div id="id2" class="group2">
</div>
```

Check box

```
<label for="id1">
    <input type="checkbox" value="class x" name="class"
                                id="id1">
</label>

<label for="id2">
    <input type="checkbox" value="class x" name="class"
                                id="id2">
</label>
```

Text area (for feedback)

```
<textarea name="feedback" id="101" placeholder="please give your
                                feedback"
```

Select

```
<select name="city" id="city">
    <option value="Delhi">Delhi</option>
    <option value="Mumbai">Mumbai</option>
    <option value="Banglore">Delhi</option>
```

</select>

iframe Tag — website inside website

<iframe src="Link">

↳ It is used to put website inside website or to show a preview of a video.

Video Tag

<video src="myVid.mp4">My Video</video>

eg –

<video width="560" height="315" src="/myVideo.mov" controls loop>my video</video>

controls — for seeing controls
loop — after finished watching this video it will loops
controls autoplay — it will used to autoplay the video.